


**DEPARTMENT OF INFORMATION TECHNOLOGY**

**AY:2022-23 (SEMESTER – I)**

**ITSE102- Introduction to Operating Systems**

**Assignment**

| | |
|---|---|
| **Student ID** | **-----** |
| **Name** | **-----** |
| **Note: 15th December is the last day to submit your assignment.** | |

**Q1. Consider the process A, B, C, D and E are arrived at the same time with the following burst time in seconds. Draw the Gantt Chart and find the average waiting time using Round Robin Scheduling. The time slice given for each process is 3 seconds.**

| PROCESS | BURST TIME |
|---|---|
| A | 15 = 3 3 3 3 3 |
| B | 17= 3 3 3 3 3 2 |
| C | 1= 1 |
| D | 11= 3 3 3 2 |
| E | 7= 3 3 1 |

## ROUND ROBIN (RR) SCHEDULING

| A | B | C | D | E | A | B | D | E | A | B | D | E | A | B | D | A | B | B |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

0 3 6 7 10 13 16 19 22 25 28 31 34 35 38 41 43 46 49 51

Waiting Time = (Turnaround time –Burst Time)

**Waiting time for process A** = 46 – 15 = 31s

(0-0) + (13-3) +(25-16) +(35-28) +(43-38) = 0+10+9+7+5=31s

**Waiting time for process B** = 51 - 17 = 34s

(3-0) + (16-6) +(28-19) +(38-31) +(46-41) +(49-49) =3+10+9+7+5=34s

**Waiting time for process C** = 7 - 1 = 6s

**Waiting time for process D** = 43 - 11 = 32s

**Waiting time for process E** = 35 - 7 = 28s

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**The average waiting time** = (31 + 34 + 6 + 32 + 28) / 5s

= 131 /5s

= 26.2s

**Turnaround time** = (46 + 51 + 7 + 43 +35 )/5

= 182/5 = 36.4s

**Q2. Consider the process A, B, C, D and E are arrived at the same time with the following burst time in seconds. Draw the Gantt Chart and find the average waiting time and average Tur Around time using Shortest Job First.**

| PROCESS | Arrival Time | BURST TIME |
|---|---|---|
| A | 1 | 15 |
| B | 1 | 17 |
| C | 0 | 1 |
| D | 3 | 11 |
| E | 7 | 7 |




Waiting time for process=(Turnaround Time – Burst Time)

Waiting time for process A = 18 s

Waiting time for process B = 33s

Waiting time for process C = 0s

Waiting time for process D = 0s

Waiting time for process E = 7s

The average waiting time = (18 +33 +0 +0 +7 ) /5
= 58/ 5s = 11.6s

Turnaround time = (Completion Time – Arrival Time)

A=34-1=33s B=51-1=50s C=1-0=1s D=14-3=11s E=21-7=14s

Turnaround time = (33 +50 +1 +11 +14) / 5s

= 109/5 = 21.8s

**Q3. Use Banker's algorithm to test if the system is in safe state or not. If the system is in safe state, write the safe sequence. Also write the NEED matrix.**

| Processes | Allocation<br>A B C | Max<br>A B C | Available<br>A B C |
|---|---|---|---|
| $P_0$ | 1 0 0 | 2 1 1 | |
| $P_1$ | 1 0 3 | 2 2 3 | 2 2 2 |
| $P_2$ | 1 1 2 | 2 1 3 | |
| $P_3$ | 1 1 1 | 3 1 2 | |
| $P_4$ | 2 1 2 | 2 3 2 | |

Solution:

1. Content of the need matrix can be calculated by using the below formula

Need = Max – Allocation

| Process | Need | | |
|---|---|---|---|
| | A | B | C |
| P0 | 1 | 1 | 1 |
| P1 | 1 | 2 | 0 |
| P2 | 1 | 0 | 1 |
| P3 | 2 | 0 | 1 |
| P4 | 0 | 2 | 0 |

**Safe sequence:**

For process P0, Need = (1, 1, 1) and

Available = (2, 2, 2)

Need ≤ Available = True

Request of P0 is granted.

∴ Available = Available +Allocation

= (2, 2, 2) + (1, 0, 0)

= (3, 2, 2) (New Available)

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For process P1, Need = (1, 2, 0) and

Available = (3, 2, 2)

Need ≤ Available = True

Request of P1 is granted.

∴ Available = Available +Allocation

= (3, 2, 2) + (1, 0, 3)

= (4, 2, 5) (New Available)

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For process P2, Need = (1, 0, 1) and

Available = (4, 2, 5)

Need ≤ Available = True

Request of P2 is granted.

∴ Available = Available +Allocation

= (4, 2, 5) +(1, 1, 2)

= (5, 3, 7) (New Available)

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For process P3, Need = (2, 0, 1) and

Available = (5, 3, 7)

Need ≤ Available = True

Request of P3 is granted.

∴ Available = Available +Allocation

= (5, 3, 7) + (1, 1, 1)

= (6, 4, 8) (New Available)

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For process P4, Need = (0, 2, 0) and

Available = (6, 4, 8)

Need ≤ Available = True

Request of P3 is granted.

∴ Available = Available +Allocation

= (6, 4, 8) + (2, 1, 2)

= (8, 5, 2) (New Available)